

Enterprise Banking Office Network

Computer Networks Project Report

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1. Executive Summary

This project presents the design and implementation of a secure banking network connecting three locations: Headquarters (HQ), Islamic Branch, and Conventional Branch. The main objective was to provide communication between all branches while protecting sensitive banking data through proper network segmentation and security controls.

The network was designed using VLANs and VLSM subnetting to efficiently divide departments into separate logical networks. Router-on-a-Stick technology was used to enable inter-VLAN communication. OSPF dynamic routing was configured to allow automatic route sharing between all routers and branches, ensuring reliable connectivity across the banking infrastructure.

A centralized server room was implemented in Headquarters to provide DHCP, DNS, HTTP, and Email services. DHCP automatically assigns IP addresses to devices, DNS resolves domain names, and the web and email servers provide communication services across the organization. These centralized services simplify management and improve consistency throughout the network.

To provide secure internet access, Network Address Translation (NAT Overload/PAT) was configured on the edge router. NAT allows multiple internal private IP addresses to share a single public IP address when accessing external networks, improving security by hiding internal network addresses from outside users.

Security was implemented using Extended Access Control Lists (ACLs). Sensitive departments such as Finance, HR, Loan, and Security were protected from unauthorized access. ACLs allow only required communication with servers and management departments while blocking unnecessary internal access.

The final network successfully achieved secure communication, centralized services, branch connectivity, dynamic routing, controlled access between departments, and secure internet access through NAT. Testing confirmed successful DHCP allocation, DNS resolution, web access, email communication, OSPF routing, NAT functionality, and ACL enforcement.

2. Network Topology Diagram

Network Overview

The network consists of four major sections:

Headquarters (HQ)

- Finance Department (VLAN 10)
- IT Department (VLAN 20)
- HR Department (VLAN 30)
- Management Department (VLAN 40)
- Wireless Access Point (VLAN 45)

Server Room

- DHCP / DNS Server
- HTTP Server
- Email Server

Islamic Branch

- Sales Department (VLAN 60)
- Shariah Department (VLAN 70)
- Security Department (VLAN 80)
- Management Department (VLAN 90)

Conventional Branch

- Sales Department (VLAN 100)
- Loan Department (VLAN 110)
- Security Department (VLAN 120)
- Management Department (VLAN 130)

WAN Connectivity

- HQ Router ↔ ISP Router
- Islamic Router ↔ ISP Router
- Conventional Router ↔ ISP Router
- HQ Router ↔ Server Room

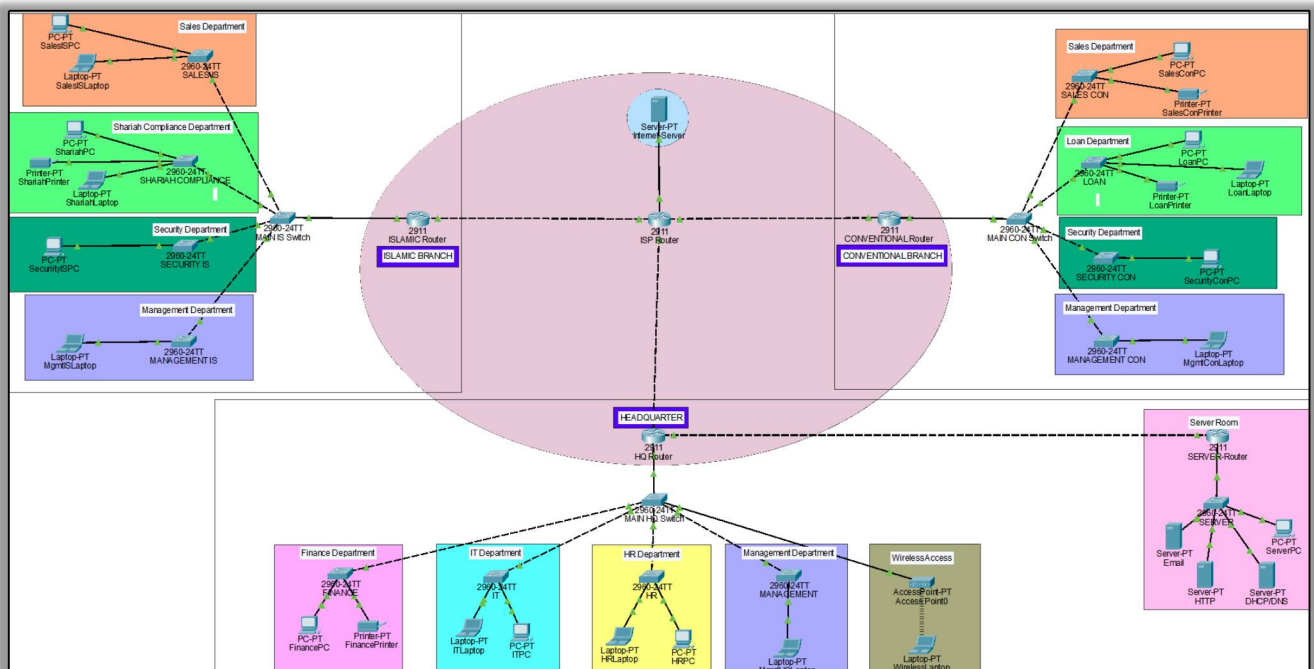


Figure 1 Complete Network Topology Diagram

3. IP Address Plan

1. HEADQUARTER

Department	Hosts Needed	VLAN ID	Network Address	Subnet Mask	Gateway	First IP	Last IP	Broadcast
Finance	20	VLAN 10	192.168.0.0/27	255.255.255.224	192.168.0.1	192.168.0.2	192.168.0.30	192.168.0.31

Department	Hosts Needed	VLAN ID	Network Address	Subnet Mask	Gateway	First IP	Last IP	Broadcast
IT	14	VLAN 20	192.168.0.32/28	255.255.255.240	192.168.0.33	192.168.0.34	192.168.0.46	192.168.0.47
HR	12	VLAN 30	192.168.0.48/28	255.255.255.240	192.168.0.49	192.168.0.50	192.168.0.62	192.168.0.63
Management	8	VLAN 40	192.168.0.64/28	255.255.255.240	192.168.0.65	192.168.0.66	192.168.0.78	192.168.0.79
Access Point	6	VLAN 45	192.168.0.80/29	255.255.255.248	192.168.0.81	192.168.0.82	192.168.0.86	192.168.0.87
Server	6	VLAN 50	192.168.0.88/28	255.255.255.240	192.168.0.89	192.168.0.90	192.168.0.102	192.168.0.103

2. ISLAMIC BRANCH

Department	Hosts Needed	VLAN ID	Network Address	Subnet Mask	Gateway	First IP	Last IP	Broadcast
Sales	12	VLAN 60	192.168.0.104/28	255.255.255.240	192.168.0.105	192.168.0.106	192.168.0.118	192.168.0.119
Shariah	7	VLAN 70	192.168.0.120/28	255.255.255.240	192.168.0.121	192.168.0.122	192.168.0.134	192.168.0.135
Security	6	VLAN 80	192.168.0.136/29	255.255.255.248	192.168.0.137	192.168.0.138	192.168.0.142	192.168.0.143
Management	5	VLAN 90	192.168.0.144/29	255.255.255.248	192.168.0.145	192.168.0.146	192.168.0.150	192.168.0.151

3. CONVENTIONAL BRANCH

Department	Hosts Needed	VLAN ID	Network Address	Subnet Mask	Gateway	First IP	Last IP	Broadcast
Sales	12	VLAN 100	192.168.0.152/28	255.255.255.240	192.168.0.153	192.168.0.154	192.168.0.166	192.168.0.167
Loan	10	VLAN 110	192.168.0.168/28	255.255.255.240	192.168.0.169	192.168.0.170	192.168.0.182	192.168.0.183
Security	6	VLAN 120	192.168.0.184/29	255.255.255.248	192.168.0.185	192.168.0.186	192.168.0.190	192.168.0.191

Department	Hosts Needed	VLAN ID	Network Address	Subnet Mask	Gateway	First IP	Last IP	Broadcast
Management	5	VLAN 130	192.168.0.192/29	255.255.255.248	192.168.0.193	192.168.0.194	192.168.0.198	192.168.0.199

4. Configuration Logs

VLAN Configuration

```

vlan 10
 name FINANCE
vlan 20
 name IT
vlan 30
 name HR
vlan 40
 name MANAGEMENT
vlan 45
 name WIRELESS

```

Trunk Configuration

```

interface g0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,45,50

```

Router-on-a-Stick Configuration

```

interface g0/1.10
 encapsulation dot1Q 10
 ip address 192.168.0.1 255.255.255.224

interface g0/1.20
 encapsulation dot1Q 20
 ip address 192.168.0.33 255.255.255.240

```

DHCP Relay

```

interface FastEthernet0/1.30
 ip helper-address 192.168.0.90

```

OSPF Routing

```

router ospf 1
 network 192.168.0.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.255 area 0

```

NAT Configuration

```
interface gigabitEthernet 0/2
ip nat outside
exit
access-list 110 permit 192.168.0.0 0.0.0.255
ip nat inside source list 110 interface g0/2 overload
```

ACL Configuration Example

```
ip access-list extended ACL_FINANCE
permit ip 192.168.0.0 0.0.0.31 192.168.0.88 0.0.0.15
permit ip 192.168.0.0 0.0.0.31 192.168.0.64 0.0.0.15
deny ip 192.168.0.0 0.0.0.31 192.168.0.0 0.0.0.255
permit ip any any
```

5. Service Configuration

DHCP Service

DHCP was used to automatically assign IP addresses, subnet masks, default gateways, and DNS server addresses to devices in all VLANs. A centralized DHCP server (192.168.0.90) was deployed in the Server Room, while IP Helper Addresses on router sub-interfaces forwarded DHCP requests from different VLANs. This eliminated manual IP configuration and simplified network management.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
CON_MGMT	192.168.0.193	192.168.0.90	192.168.0.194	255.255.255.248	6	0.0.0.0	0.0.0.0
CON_SECURITY	192.168.0.185	192.168.0.90	192.168.0.186	255.255.255.248	6	0.0.0.0	0.0.0.0
CON_LOAN	192.168.0.169	192.168.0.90	192.168.0.170	255.255.255.240	6	0.0.0.0	0.0.0.0
CON_SALES	192.168.0.153	192.168.0.90	192.168.0.154	255.255.255.240	6	0.0.0.0	0.0.0.0
IS_MGMT	192.168.0.145	192.168.0.90	192.168.0.146	255.255.255.248	6	0.0.0.0	0.0.0.0
IS_SECURITY	192.168.0.137	192.168.0.90	192.168.0.138	255.255.255.248	6	0.0.0.0	0.0.0.0
IS_SHARIAH	192.168.0.121	192.168.0.90	192.168.0.122	255.255.255.240	6	0.0.0.0	0.0.0.0
IS_SALES	192.168.0.105	192.168.0.90	192.168.0.106	255.255.255.240	6	0.0.0.0	0.0.0.0
IP_HELPER	192.168.0.94	192.168.0.90	192.168.0.95	255.255.255.248	6	0.0.0.0	0.0.0.0

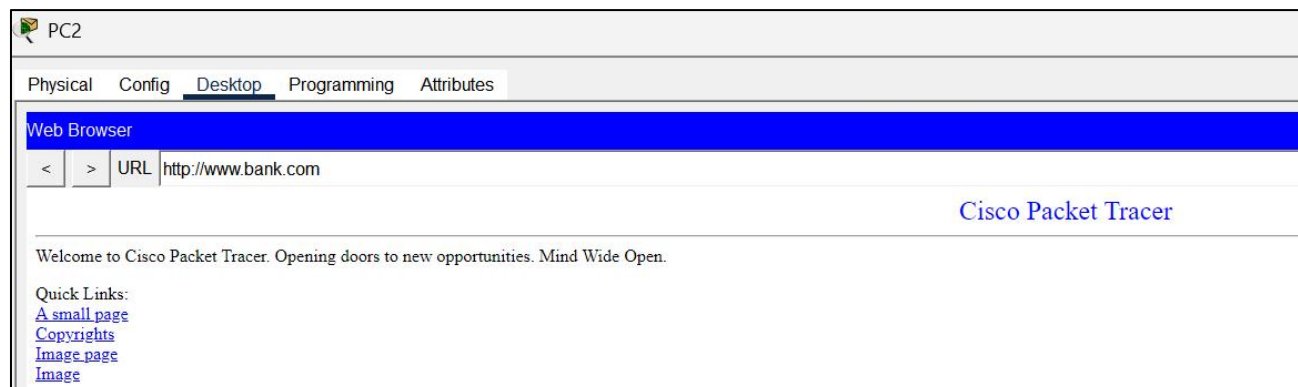
DNS Service

DNS was used to translate domain names into IP addresses. Instead of accessing services using numerical IP addresses, users could access resources through easy-to-remember names such as www.bank.com and mail.bank.com. The centralized DNS server at 192.168.0.90 provided name resolution for all branches and departments.

No.	Name	Type	Detail
0	mail.bank.com	A Record	192.168.0.92
1	www.bank.com	A Record	192.168.0.91

HTTP Service

The HTTP service was used to host the organization's internal banking website on Server1 (192.168.0.91). Users from Headquarters, Islamic Branch, and Conventional Branch accessed the website through www.bank.com to verify connectivity, DNS functionality, and web service availability across the network.

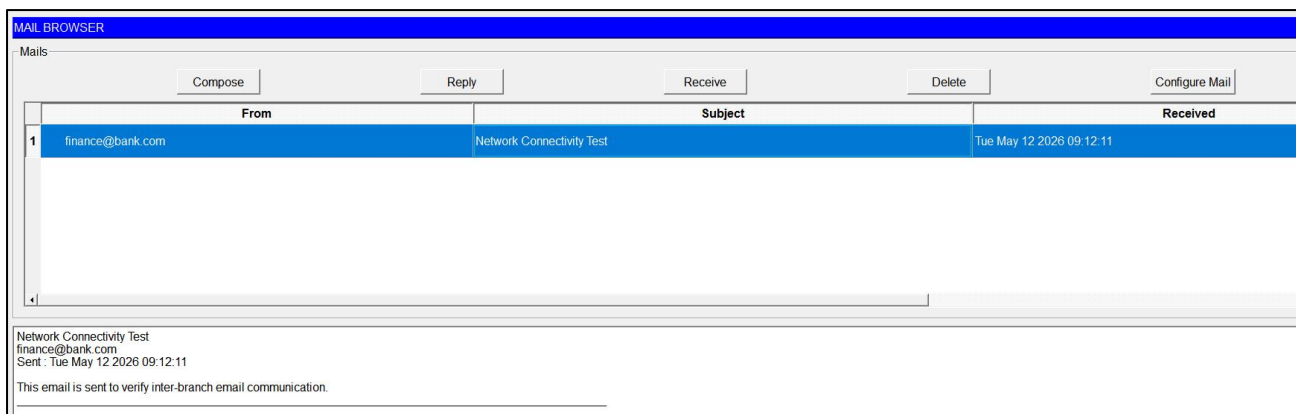


Email Service

The Email service was used to enable communication between employees located in different departments and branches. The mail server (192.168.0.92) hosted user accounts such as finance@bank.com, hr@bank.com, sales@bank.com, and loan@bank.com. Successful email exchange between branches verified proper routing, DNS resolution, and server connectivity throughout the banking network.

Username	Email	Department
finance	finance@bank.com	HQ Finance
hr	hr@bank.com	HQ HR
it	it@bank.com	HQ IT
management	management@bank.com	All Management
sales	sales@bank.com	IS & CON Sales
shariah	shariah@bank.com	IS Shariah
loan	loan@bank.com	CON Loan
security	security@bank.com	IS & CON Security

Receiving mail from POP3 Server mail.bank.com
 DNS resolving. Resolving name: mail.bank.com by querying to DNS Server: 192.168.0.90 DNS resolved ip address: 192.168.0.92
 Receive Mail Success.



6. Security Policy

The banking network contains sensitive financial and customer information. To protect this data, Extended Access Control Lists (ACLs) were configured on router sub-interfaces

Source Department	Permitted Access	Denied Access
Finance (VLAN 10)	Server Room, HQ Management, IT Department, Internet	All other internal departments
HR (VLAN 30)	Server Room, HQ Management, Internet	All other internal departments
Wireless Users (VLAN 45)	DNS Server, HTTP Server, Internet	All internal banking departments
Islamic Sales (VLAN 60)	Server Room, Internet	All other internal departments
Shariah (VLAN 70)	Server Room, HQ Management, Internet	All other internal departments
Islamic Security (VLAN 80)	Server Room, Internet	All other internal departments
Conventional Sales (VLAN 100)	Server Room, Internet	All other internal departments
Loan Department (VLAN 110)	Server Room, HQ Management, Internet	All other internal departments
Conventional Security (VLAN 120)	Server Room, Internet	All other internal departments

Justification of ACL Rules

Finance Department

Finance users are allowed to access the Server Room, IT Department, and HQ Management because these resources are required for daily banking operations and technical support. Access to all other departments is blocked to protect confidential financial records.

HR Department

HR users can access the Server Room and HQ Management for payroll and employee management tasks. Access to other departments is denied to protect employee information.

Wireless Users

Wireless users are only allowed to access DNS services, the banking website, and the Internet. Access to internal banking networks is denied to prevent unauthorized access from guest devices.

Islamic Branch Departments

Islamic Sales, Shariah, and Security departments can access centralized servers required for business operations. Access to unrelated internal departments is blocked to maintain data confidentiality and branch isolation.

Conventional Branch Departments

Conventional Sales, Loan, and Security departments can access central servers and required management networks. Access to other internal banking departments is denied to prevent unauthorized data access and reduce security risks.

7. Verification Evidence

Ping Testing

Source	Destination	Protocol	Result
Finance PC	Web Server (192.168.0.91)	Ping	Success
HR PC	Web Server (192.168.0.91)	Ping	Success
Islamic Sales PC	Web Server (192.168.0.91)	Ping	Success
Loan PC	Web Server (192.168.0.91)	Ping	Success
Finance PC	HR PC	Ping	Blocked
HR PC	Finance PC	Ping	Blocked
Finance PC	IT PC	Ping	Success
Shariah PC	HQ Management PC	Ping	Success

Traceroute Commands

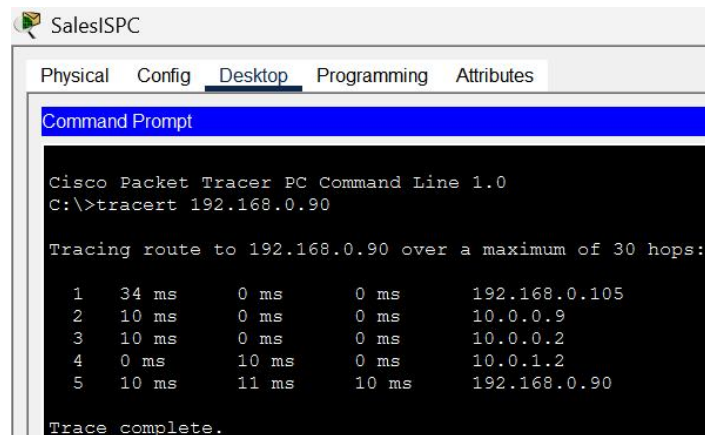


Figure 2 Sales to Server

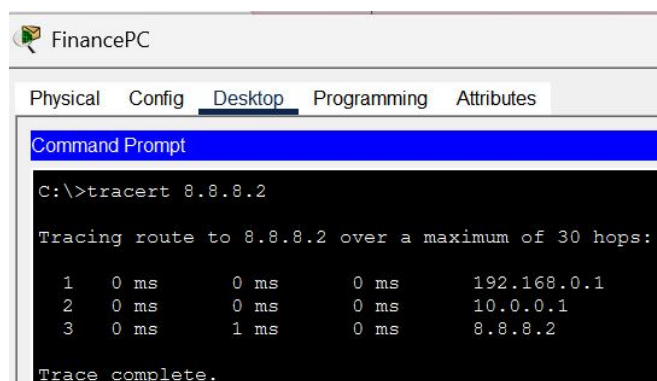


Figure 3 Finance to Internet

Commands Outputs

show ip ospf neighbor

```

HQ_Router>show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.0.89   1     FULL/DR         00:00:32   10.0.1.2       GigabitEthernet0/2
192.168.0.185  1     FULL/DR         00:00:32   10.0.0.1       GigabitEthernet0/0

```

show access-lists

```

HQ_Router#show access-lists
Extended IP access list 110
 10 permit ip 192.168.0.0 0.0.0.31 host 8.8.8.2
 20 permit ip 192.168.0.32 0.0.0.15 host 8.8.8.2
 30 permit ip 192.168.0.48 0.0.0.15 host 8.8.8.2
 40 permit ip 192.168.0.64 0.0.0.15 host 8.8.8.2
 50 permit ip 192.168.0.80 0.0.0.7 host 8.8.8.2
Extended IP access list ACL_FINANCE
 10 permit ip 192.168.0.0 0.0.0.31 192.168.0.80 0.0.0.15 (70 match(es))
 20 permit ip 192.168.0.0 0.0.0.31 192.168.0.64 0.0.0.15
 30 permit ip 192.168.0.0 0.0.0.31 192.168.0.32 0.0.0.15 (4 match(es))
 40 deny ip 192.168.0.0 0.0.0.31 192.168.0.0 0.0.0.255
 50 permit ip any any (5 match(es))
Extended IP access list ACL_HR
 10 permit ip 192.168.0.48 0.0.0.15 192.168.0.80 0.0.0.15 (1 match(es))
 20 permit ip 192.168.0.48 0.0.0.15 192.168.0.64 0.0.0.15
 30 deny ip 192.168.0.48 0.0.0.15 192.168.0.0 0.0.0.255
 40 permit ip any any (4 match(es))
Extended IP access list ACL_WIRELESS
 10 permit udp 192.168.0.80 0.0.0.7 host 192.168.0.90 eq domain
 20 permit tcp 192.168.0.80 0.0.0.7 host 192.168.0.90 eq www
 30 deny ip 192.168.0.80 0.0.0.7 192.168.0.0 0.0.0.255
 40 permit ip any any

```

show ip route

```

HQ_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
O       10.0.0.4/30 [110/2] via 10.0.0.1, 00:47:11, GigabitEthernet0/0
O       10.0.0.8/30 [110/2] via 10.0.0.1, 00:47:11, GigabitEthernet0/0
C       10.0.1.0/30 is directly connected, GigabitEthernet0/2
L       10.0.1.1/32 is directly connected, GigabitEthernet0/2
    192.168.0.0/24 is variably subnetted, 18 subnets, 4 masks
C       192.168.0.0/27 is directly connected, GigabitEthernet0/1.10
L       192.168.0.1/32 is directly connected, GigabitEthernet0/1.10
C       192.168.0.32/28 is directly connected, GigabitEthernet0/1.20
L       192.168.0.33/32 is directly connected, GigabitEthernet0/1.20

```

show ip nat translations

```

HQ_Router>show ip nat translations
Pro  Inside global      Inside local      Outside local     Outside global
icmp 10.0.0.2:27          192.168.0.3:27   8.8.8.2:27       8.8.8.2:27
icmp 10.0.0.2:28          192.168.0.3:28   8.8.8.2:28       8.8.8.2:28
icmp 10.0.0.2:29          192.168.0.3:29   8.8.8.2:29       8.8.8.2:29
icmp 10.0.0.2:30          192.168.0.3:30   8.8.8.2:30       8.8.8.2:30

```

Successful Ping Tests

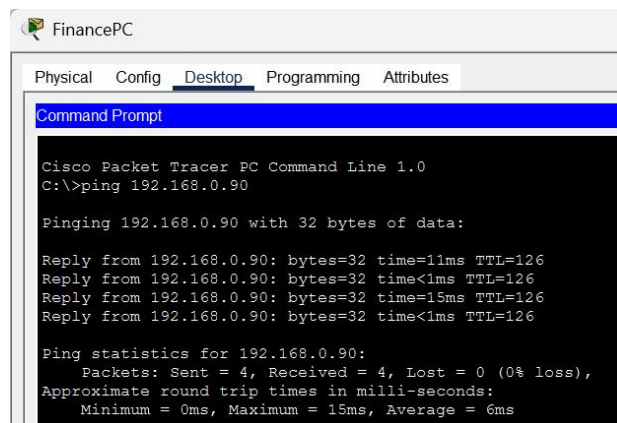


Figure 4 Finance to DHCP Server

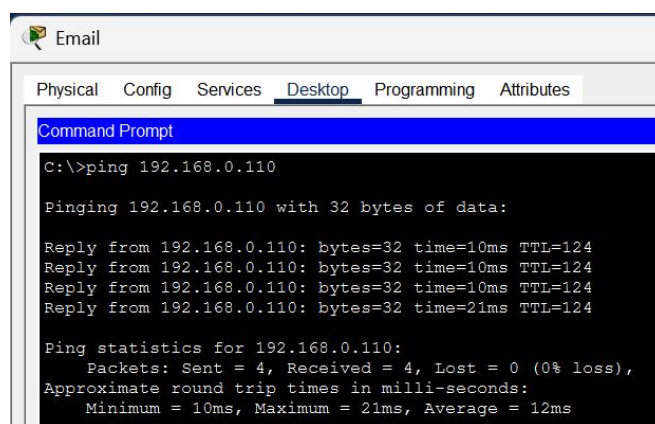


Figure 5 Email Server to SalesIS_PC

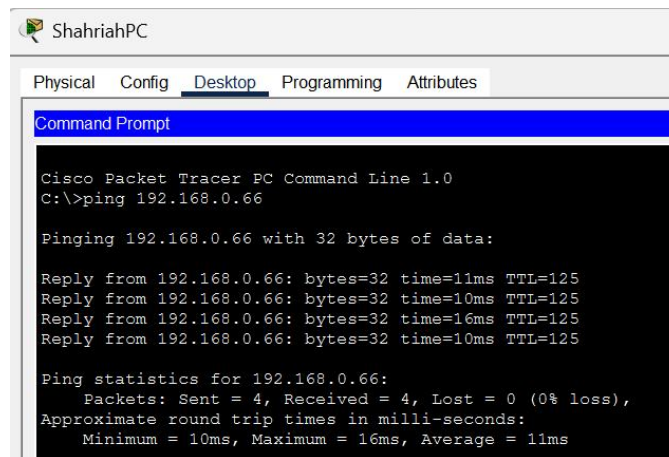


Figure 6 Shariah to HQ_Mgmt

Failed Ping Tests

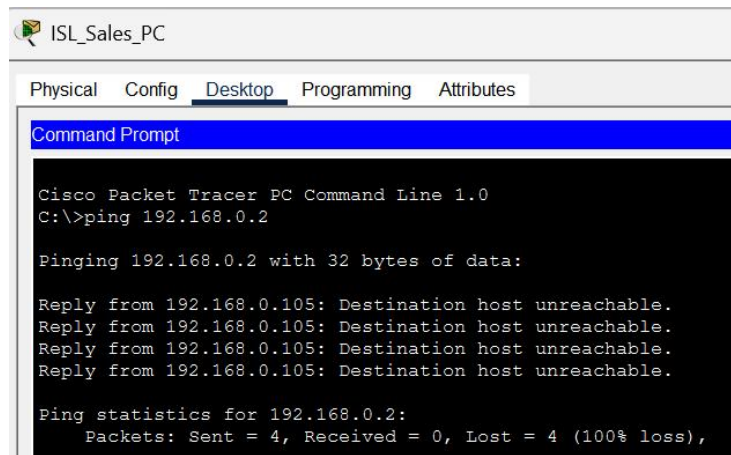


Figure 7 ISL_Sales to Finance

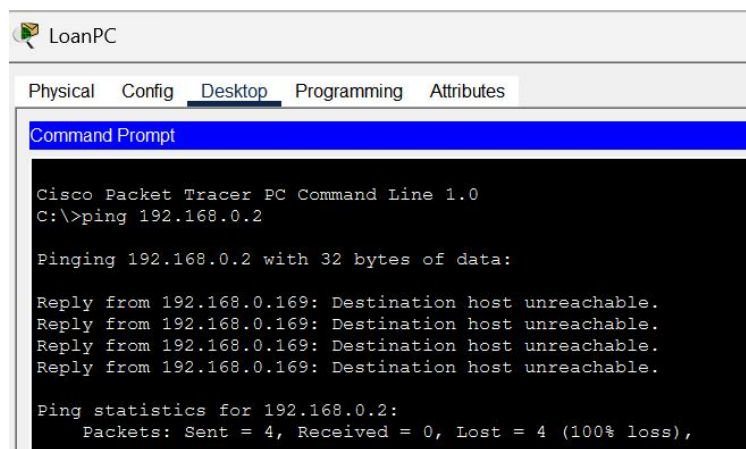


Figure 8 Loan to Finance

8. Challenges and Solutions

Challenge 1: Inter-VLAN Communication Failure

Problem: Initially, devices located in different VLANs were unable to communicate with each other. Users in separate departments could only communicate within their own VLAN, preventing access to centralized services and resources.

Resolution: Router-on-a-Stick inter-VLAN routing was implemented by creating sub-interfaces for each VLAN on the branch routers. IEEE 802.1Q encapsulation and gateway IP addresses were configured, allowing controlled communication between VLANs.

Challenge 2: Missing OSPF Route Advertisements

Problem: Branch devices were unable to access servers located at Headquarters because routing information was not being exchanged correctly between routers.

Resolution: OSPF Process 1 was configured on all routers, and the correct WAN and VLAN networks were advertised. Once OSPF neighbors reached the FULL state, all routes were learned dynamically and connectivity was restored.

Challenge 3: ACL Rules Not Enforcing Security Policies

Problem: After applying ACLs, some departments could still communicate with unauthorized internal networks. The ACL configuration was not producing the expected security restrictions.

Resolution: The ACL rule order was reviewed and corrected. Specific permit statements were placed before deny statements, followed by a final permit statement for Internet access. This ensured that only authorized traffic reached internal resources.

9. Conclusion

This project successfully implemented a secure banking network connecting Headquarters (HQ), the Islamic Branch, and the Conventional Branch. VLANs, VLSM, OSPF, DHCP, DNS, HTTP, Email services, NAT, and ACLs were configured to provide secure communication, centralized services, and controlled access between departments and branches.

Testing confirmed successful routing, IP address assignment, DNS resolution, web access, email communication, Internet connectivity, and ACL enforcement. The network met its functional and security objectives.

Future Improvements

- Firewall Appliances
- VPN Connectivity
- IDS/IPS Solutions
- Redundant ISP Connections
- Server Backup and Redundancy
- Network Monitoring Tools
- Multi-Factor Authentication (MFA)

These additions would improve security, reliability, and availability in a real banking environment.
